

Reproducible environment and records

Project: Reproducible AI Research Study

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Power and resource tradeoffs

Without opening the old lesson, explain one key idea from Power and resource tradeoffs. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Predefining an analysis plan

Without opening the old lesson, explain one key idea from Predefining an analysis plan. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Dataset and benchmark limitations

Without opening the old lesson, explain one key idea from Dataset and benchmark limitations. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Reproducible environment and records. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Reproducible environment and records. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Partner changes preprocessing and data version while keeping old run ID. What must be preserved?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Folder contains seed42 and final-score.txt, but no dataset, code or dependency record. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: Reproducibility supports checking the computation, not automatic validity of the research claim. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Partner changes preprocessing and data version while keeping old run ID. What must be preserved?

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled reproducible environment and records evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of reproducible environment and records. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Reproducible environment and records

A reproducible calculation record. A teammate must rerun a three-value mean without guessing the input or software version.

Data file contains 2,4,6. Recorded script sums values and divides by count; environment and file hash are saved.

Recorded mean calculation

	Value 1	Value 2	Value 3	Mean
Original data	2	4	6	4
Changed data	2	4	9	5

Read a cell using both its row and column labels.

Reproducible environment and records

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Identify exact input

Use the three recorded values, not a similarly named edited file.



2. Trace calculation

Sum is 12; count is 3; mean is 4.



3. Save provenance

Retain script version, environment description and data hash with the output.

CHANGE ONE THING

The data file changes to 2,4,9.

Explain the changed result using the matching lesson.

Trace the recorded mean for 2,4,6 and its changed result for 2,4,9. Which artefacts must accompany the output so another team can identify the exact run?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

13. Project portfolio milestone

Project: Reproducible AI Research Study

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A reproducible calculation record. A teammate must rerun a three-value mean without guessing the input or software version.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Retain script version, environment description and data hash with the output.

1. Trace calculation
2. Identify exact input
3. Save provenance

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Save provenance Retain script version, environment description and data hash with the output. Reproducibility needs the materials and decisions that produced a result. A screenshot of 4 does not specify which data or calculation generated it.

Practice 2: explain the deciding evidence

Changed-case practice: The data file changes to 2,4,9. Which conclusion is supported by the supplied facts?

1. The mean becomes $15/3=5$, and the saved input hash must change.
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The mean becomes $15/3=5$, and the saved input hash must change. The calculation is the same but its input records differ.

Repair a missing building block

Review Power and resource tradeoffs (ai-7/23). Goal: Explain the mechanism and terminology of power and resource tradeoffs. Short practice: Calculate both simulated detection fractions and resource costs. What additional detection is observed for the extra forty units, and does that establish power for an actual classroom study?. Return to this week and explain what you changed.

Review Predefining an analysis plan (ai-7/21). Goal: Explain the mechanism and terminology of predefining an analysis plan. Short practice: Apply the recorded exclusion rule. Can two readable model failures be removed, and how must a necessary later plan change be disclosed?. Return to this week and explain what you changed.

Review Dataset and benchmark limitations (ai-7/18). Goal: Explain the mechanism and terminology of dataset and benchmark limitations. Short practice: Report the indoor benchmark with its denominator and missing condition. After two correct outdoor cases out of four are added, what slice rate is supported and what remains uncertain?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

